



DAVID ÁLVAREZ ROSA

Mathematician · Engineer · Dublin



Passionate about low-latency, high-performance systems. Currently in algorithmic trading at Susquehanna; previously at Amazon, Sopra Steria, and Deloitte. Holds a BSc in Mathematics, a BEng in Industrial Engineering, and an MSc in Artificial Intelligence.

EXPERIENCE

Software Engineer Jul 2024–Present
Susquehanna International Group · Dublin, Ireland

High-frequency options trading. Low-latency market data and trading signals. Mentor and interviewer.

Software Engineer II Mar 2022–Aug 2024
Amazon · Madrid, Spain

Designed systems for 10M+ monthly active customers. Contributed to 100+ internal repos. Won org hackathon. Promoted in 18 months (top 5%). Mentored 3. On-call. Interviewer.

Machine Learning Engineer Apr 2024–Jul 2024
Sopra Steria · Remote

Researched, designed, and built a semantic cache for LLMs.

Risk Analyst Sep 2021–Mar 2022
Deloitte · Madrid, Spain

Quantitative analysis of technological and cybersecurity risks for top-tier banking companies.

Visiting Researcher Sep 2020–Jun 2021
Vector Institute · Toronto, Canada

Research thesis on multimodal learning (recomprehension.com).

Machine Learning Engineer Sep 2019–Feb 2020
BCN eMotorsport · Barcelona, Spain

Perception at Driverless UPC. Served as LiDAR lead and collaborated on computer vision for a fully autonomous car.

EDUCATION

MSc in Artificial Intelligence 9.00/10
Official study program focused on AI research and enabling PhD. Honors in 6 subjects.

MSc in Mathematics
Math-lover part-time student. Dropout (joined Amazon).

Research Thesis 10/10 (A+)
Research thesis on multimodal learning at University of Toronto.

BSc in Mathematics 8.12/10 (top 10%)
Rigorous and proof-oriented degree with a robust mathematical base. Honors in 9 subjects.

BEng in Industrial Engineering 8.03/10 (top 2%)
Multidisciplinary and integrative vision of industrial engineering. Honors in 14 subjects.

CERTIFICATIONS

Advanced English (C1)—Cambridge
Machine Learning—Stanford University
Deep Learning—deeplearning.ai
Blockchain—Hong Kong University
Nova Talent Member—Nova

HONORS & AWARDS

Mathematical Olympiad
Silver (local), honors (national).

Physics Olympiad
Gold (local), silver (national).

Mobility Scholarship
Research thesis at Toronto (€6k).

Tuition and Housing Scholarship
University tuition and housing (€19k).

General Scholarship
Full tuition plus annual stipend (€11k).

VOLUNTEERING

Mathematics Tutor
Training for the Mathematical Olympiads.

Volunteer @ Banco de Alimentos
Food collection for people in need.

LANGUAGES

English—Proficient
Spanish—Native
Catalan—Intermediate

CONTACT

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